CSCI323: Modern Artificial Intelligence
2025 Autumn Session

Question 1 (5 Marks) You have a loss function:

$$\text{Loss}(x, y, z, w) = 1.5(x^2y + \max\{w, z\}). \qquad (1)$$

Use backpropagation algorithm to compute the gradients for the four variables at x = 4, y = -2, z = 5 and w = -1. Provide the computation graph with the following nodes: addition, square, multiplication, max, multiplication by a constant.

Question 2 (5 Marks) Assume you have two datasets of (x, y):

- $D_1 = \{(-1, +1), (0, -1), (1, +1)\}$.
- $D_2 = \{(-1, -1), (0, +1), (1, -1)\}$.

If you use feature $\varphi(x) = x$, neither dataset can be linearly separated. You need to define a two-dimensional feature $\varphi(x)$ to fix this such that:

- A weight vector $\mathbf{w}_1$ can classify D_1 perfectly (i.e., $\mathbf{w}_1 \cdot \varphi(x) > 0$ if x has label +1 and $\mathbf{w}_1 \cdot \varphi(x) < 0$ if x has label −1); and
- A weight vector $\mathbf{w}_2$ can classify D_2 perfectly;

Note the two datasets share the same features $\varphi(x)$ but the weight vectors can be different. $\varphi(x) = x$ normally can be re-written as $\varphi(x) = [1, x]$, which is regarded as one-dimensional feature.

Question 3 (5 Marks) Suppose you want to travel from city 1 to city n (going only forward) and then travel back to city 1 (going only backward). The cost to go from place i to j is $c_{i,j} \geq 0$. Select all the algorithms from the following algorithms that can be used to find the minimum cost path. Present the reasons why an algorithm applies or does not apply.

- a. depth-first search
- b. breadth-first search
- c. dynamic programming
- d. uniform cost search

Submission:
Due date: **18 April 2025 (Friday in Session Week 7), 11:30pm**.
Submit a PDF file to answer all the questions. Structure your answers: paragraphs, bullet points, etc.

Late submission:
5% deduction per day. Submissions received **7 days** after the due date will receive no marks.

Plagiarism:
A plagiarized assignment will receive a zero mark and be penalized according to the university rules. Plagiarism detection software might be used for this assignment.